PS

PadhSang

A secure collaborative workspace for student projects

PLAN. COLLABORATE. PROGRESS.

Deepesh Kumar • SID 2327937 • Supervisor: Priyanka Sharma

I built, tested and investigated PadhSang to answer how third-party authentication and server-side authorisation can support secure student collaboration.

The tested system passed 25/25 automated checks and 5/5 complete browser workflows.

Problem & aim

Student teams often split discussions, deadlines and personal planning across different tools.

Aim: design and evaluate one secure workspace for project collaboration and focused study.

Research question: How can OAuth and server-side access control protect a practical student workspace?

What I built

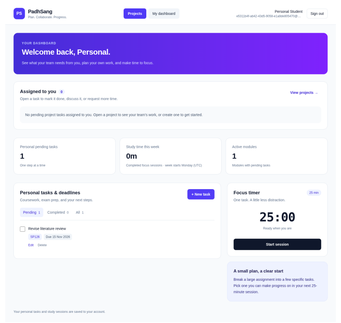
- Google and Microsoft OAuth support
- Project invitations and membership
- Assigned tasks, priorities and deadlines
- Extension requests and owner decisions
- Discussion and task feedback
- In-app reminders and private planner
- Persistent 25-minute focus timer

Secure architecture

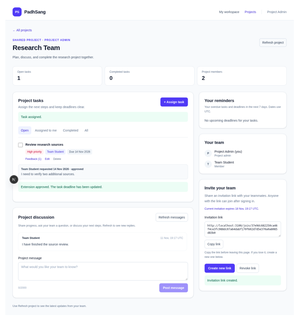
Student → Google/Microsoft OAuth → Next.js/Auth.js server checks → Prisma/PostgreSQL

Membership and ownership are checked on protected operations. Invitation tokens are random, hashed and expire after seven days.

Application evidence



Personal dashboard: assigned work, planner and focus timer



Project workspace: permissions, task workflow and discussion

Testing & results

25/25	5/5	12
SERVICE + SECURITY TESTS	BROWSER WORKFLOWS	DATABASE TABLES

Covered data isolation, task ownership, project membership, invitations, extension decisions, feedback, timer persistence and responsive layouts.

Production build passed • PostgreSQL migration passed

What I found

Testing indicated that the implemented server-side checks separated personal and project data in the tested scenarios.

Members completed assigned workflows, while non-members were denied project access.

Security also depended on external OAuth, database and hosting configuration—not only application code.

Security decisions

- Database-backed authenticated sessions
- Owner/member/assignee permission checks
- Hashed, expiring invitation tokens
- Transactional extension decisions
- Safe internal redirect validation
- No production test-login bypass

Limitations

- Microsoft live sign-in still needs provider credentials and real-account verification.
- Discussion refresh is manual; reminders are in-app only.
- Automated checks are not a formal penetration test or user study.
- Initial hosting was suspended pending provider review.

Conclusion

PadhSang demonstrates that secure authentication, role-based access and collaborative study tools can be combined in one practical student workspace.

Source: github.com/deepeshkumarbangrwa-sys/study-sync • Evidence uses isolated synthetic test accounts • References: OWASP; Auth.js; Next.js; Prisma